4.Workflow Automation

One of the most important parts of the system is the **workflow automation** that manages the book borrowing process. ServiceNow' s **Flow Designer** is used to automate actions when specific events occur in the system.

When a student submits a borrow request, the request enters the workflow and waits for librarian approval. Once the librarian reviews and approves the request, the workflow automatically updates the book status to " **Issued."**

This automation eliminates the need for librarians to manually update book records, reducing errors and saving time.

Example Workflow Process

1. A student submits a borrow request through the portal.
2. The request is stored in the **Borrow Request Table** with the status **Pending**.
3. The librarian reviews the request in the system.
4. If the librarian approves the request, the **Flow Designer** workflow automatically updates the status to **Approved**.
5. The book status in the **Book Table** changes from **Available** to **Issued**.

Example Scenario

A student named **Anita** requests the book " *Operating Systems Concepts."* The librarian approves the request. Immediately after approval, the Flow Designer automatically updates the book' s status to **Issued**, ensuring that other students cannot request the same book until it is returned.

Business Rules and Notifications

To improve communication between the system and its users, **Business Rules and Email Notifications** are implemented. Business rules

automatically execute specific actions when certain conditions are met.

In this project, a business rule triggers an **email notification** whenever a borrow request is approved or rejected. This ensures that students receive real-time updates about the status of their requests.

Notification Process

1. The librarian approves or rejects a borrow request.
2. A **Business Rule** detects the change in request status.
3. The system automatically sends an email notification to the student.
4. The student receives the message informing them about the request status.

Example Email Notification

If a student's request is approved, they may receive an email like this:

Subject: Book Request Approved

Message:
"Your request for the book *Database Management Systems* has been approved. Please visit the library to collect the book."

If the request is rejected, the system sends a message explaining the reason, such as the book being unavailable.

Example Scenario

A student named **Karthik** requests the book "*Computer Networks.*" The librarian approves the request. Immediately after approval, the system sends Karthik an email confirming that the request has been approved and the book is ready for collection.

This automated notification system improves communication and reduces confusion for users.

Reference Qualifiers

Reference Qualifiers are implemented to improve the accuracy of data entry when students submit borrow requests. A reference qualifier filters the available options in a reference field based on specific conditions.

In this project, reference qualifiers ensure that students can only select books that are currently **Available** in the library.

This prevents students from requesting books that are already issued or unavailable.

Example Process

When a student opens the borrow request form, the **Book selection field** only displays books with the status **Available**. Books that are already borrowed or reserved are hidden from the list.